

1 Humanitarian Giving Observatory

1.1 Project overview

Humanitarian Giving Observatory is a Power BI portfolio dashboard that examines how reported funding compares with humanitarian funding requirements across reporting years and locations. It is designed as a concise, recruiter-ready financial observatory rather than as a live operational system.

The governing question is:

How well does reported funding cover global humanitarian funding requirements?

The Executive Overview answers this broad question through four KPI cards, an annual requirements-and-coverage combo chart, a geographic coverage Shape Map, and a Top 10 reported funding-gap ranking. The dashboard describes reported financial requirements and funding coverage. It must not be interpreted as a direct measure of humanitarian need, severity, affected people, or human suffering.

The project uses public data from the [United Nations Office for the Coordination of Humanitarian Affairs Financial Tracking Service](#) and is built for cleaning, modeling, analysis, and dashboard communication in Microsoft Power BI.^{1 2}

1.2 Project status and scope

The project has been narrowed to a requirements-only analytical model. The incoming- and outgoing-funding queries were removed from the active model because they are not required to answer the final Executive Overview question. The active analysis is based on the requirements-and-funding extract, supported by a country-code lookup.

The Power BI report contains the Executive Overview and may retain a concise Data & Methods page as an in-report reference. The README remains the authoritative technical document for data provenance, transformations, model logic, measures, validation, and limitations.

Scope item	Final treatment
Requirements	Retained as the primary financial requirement measure
Reported funding	Retained as the funding recorded against requirements
Funding gap	Calculated as the requirement shortfall; non-shortfall contexts display zero in the shortfall view
Coverage	Calculated as reported funding divided by requirements
Location	Country-code dimension enriched with country name, ISO Alpha-3, region, and sub-region
Time	Annual reporting_year; no daily dim_date is required for the current dashboard

Scope item	Final treatment
Incoming and outgoing funding	Removed from the active requirements-only model

1.3 Data sources

The core source file is `fts_requirements_funding_global.csv`, an OCHA FTS global requirements-and-funding extract. The country lookup is maintained separately to translate cleaned location codes into readable and mappable geographic attributes.

File or query	Role in the final model
<code>fts_requirements_funding_global.csv</code>	Core source for annual requirements, reported funding, and source coverage values
<code>stage requirement funding</code>	Staging query that preserves the source structure before analytical transformations
<code>fact requirement funding</code>	Requirements fact table used by the measures and visuals
<code>dim requirement plan</code>	Plan dimension containing the country-plan key and plan attributes
<code>dim requirement location</code>	Country/location dimension containing cleaned codes and lookup attributes
<code>stage country code ISO three</code>	Staging version of the country-code reference
<code>lookup country code ISO three</code>	Enrichment lookup containing <code>location_code</code> , <code>country_name</code> , <code>iso_alpha_3</code> , <code>region</code> , and <code>sub_region</code>

The primary public source is the [HDX Global Requirements and Funding Data page](#).¹ OCHA describes FTS as a system for tracking and publishing humanitarian funding information.² The country-code reference is based on the [DataHub country-code CSV](#).³

1.4 Coverage and analytical grain

The downloaded requirements file contains annual records covering reporting years from **1999 through 2031**. The current dashboard view is focused on the recorded annual reporting period used in the Executive Overview, with the visible trend spanning approximately **2000–2026**. Future or incomplete years must be interpreted cautiously because a low coverage value may reflect reporting incompleteness rather than a settled annual outcome.

The analytical grain of the requirements fact is one **country-plan-reporting-year record**, with country/year records also possible where the source does not provide a named plan. The `country_plan_key` combines the cleaned country code and plan identifier so that the fact can connect to the appropriate plan row. This key is retained because the fact is not simply a country-level table; multiple plans can exist for the same country and reporting year.

The source fields used by the requirements staging query include `country_code`, `id`, `name`, `code`, `type_id`, `type_name`, `start_date`, `end_date`, `year`, `requirement`, `funding`, and `percent_funded`. The analytical fact standardises the key fields as the reporting year, requirement amount, reported-funding amount, source coverage value where retained, and the country-plan key.

1.5 Cleaning and transformation approach

The source is first preserved in a staging query. The analytical fact is then created as a separate reference query so that source columns and transformation steps remain traceable. Code fields are treated as text to preserve their identifiers, numeric measures are typed as numeric values, and annual reporting fields are typed as whole numbers.

The location dimension is rebuilt from the requirements staging query rather than from the deleted incoming or outgoing funding sources. The `country_code` field is trimmed, cleaned, and converted to uppercase. Blank or null values were not present in the rebuilt location reference, and duplicate country codes were removed so that the dimension contains one row per unique country code.

The cleaned location query is merged with `lookup country code ISO three` using the location code. The expanded fields are `country_name`, `iso_alpha_3`, `region`, and `sub_region`. The ISO Alpha-3 field is retained for geographic consistency, while the current Shape Map uses the readable country field with the corresponding country attributes available for mapping and tooltips.

The plan dimension is built at the country-plan grain. The `country_plan_key` is the explicit business key used to connect plan records to the requirements fact. An independent Power Query index is not used as a substitute for this matching logic; if a numeric surrogate were ever added, it would have to be propagated by matching the existing stable business key.

The source `percent_funded` value is not summed or averaged to create the headline dashboard coverage. Coverage is recalculated from the additive requirement and reported-funding measures so that the result remains coherent under year, plan, and location filters.

1.6 Final data model

The final model is a requirements-focused **snowflake schema**. The location dimension filters the plan dimension through the country code, and the plan dimension filters the requirements fact through the country-plan key.

```
dim requirement location[country_code]
    1 → *
dim requirement plan[country_code]
    1 → *
fact requirement funding[country_plan_key]
```

The two relationships are active, one-to-many, and single-directional. Staging queries and the source lookup are reference-only and are not used as report-model relationship endpoints.

This structure is appropriate because the fact table is recorded at the country-plan-reporting-year grain, while the location dimension is intentionally deduplicated to one row per country. A direct location-to-fact relationship would require `country_code` to be added to the fact and would create a different, flatter

star-schema design. The current model avoids that additional denormalisation and preserves the existing plan key logic.

1.7 Core measures

The dashboard uses four core measures. Their definitions are deliberately simple and remain consistent across KPI cards, the annual chart, the gap ranking, and the coverage map.

Measure	Definition	Interpretation
Requirements_Total	Sum of valid requirement amounts	Recorded financial requirements in the selected context
Requirements_Reported_Funding	Sum of valid reported-funding amounts	Funding reported against those requirements
Funding_Gap_Shortfall	Positive part of requirements minus reported funding	Reported shortfall; zero when reported funding meets or exceeds requirements
Coverage_Percent	Reported funding divided by requirements	Requirements-weighted reported funding coverage

The coverage calculation is:

```
Coverage_Percent =  
DIVIDE (  
    [Requirements_Reported_Funding],  
    [Requirements_Total]  
)
```

Coverage is not capped at 100%. Values above 100% indicate that reported funding exceeds the recorded requirement in the selected context. They are financial reporting signals, not measures of humanitarian outcomes.

1.8 Executive Overview logic

The four KPI cards provide the immediate financial summary. Under the current validation context, with the report filters cleared or set to the confirmed all-period state, they display approximately \$531.38bn in requirements, \$430.83bn in reported funding, \$100.55bn in funding gap, and 81.08% coverage. These values are validation checks under a filter context and must not be treated as immutable global totals when a year, location, or plan filter is active.

The annual combo chart uses `reporting_year` on the horizontal axis. Annual requirements and annual reported funding are shown as clustered columns, while `Coverage_Percent` is shown as the line. Its narrative question is:

How closely has reported funding matched annual humanitarian requirements over time?

The Shape Map uses the cleaned location dimension and coverage as its color logic. The map is explicitly a geographic coverage visual, not a ranking of total requirements. Locations are grouped into four coverage bands: below 50%, 50% to below 80%, 80% to below 100%, and 100% or more. Its narrative question is:

How does reported funding coverage vary across locations?

The gap-ranking visual shows the Top 10 largest positive shortfalls. Its purpose is to identify locations where recorded requirements exceed reported funding by the largest amount. It must not be described as a ranking of humanitarian suffering or unmet human need.

The Reporting Year slicer is used for the context-sensitive location and shortfall analysis. The annual chart remains a time-series view so that users can see the full coverage pattern across reporting years rather than reducing it to a single selected year.

1.9 Data quality signals and interpretation guardrails

The requirements file contains incomplete and non-standard reporting patterns. Requirements, reported funding, and source coverage fields should be checked for blanks and invalid values before being used in a visual. Future reporting years and partial years require a completeness caveat. Coverage may exceed 100%, and that result should remain visible rather than being capped or treated as an error.

The source describes reported financial records. It does not measure the full universe of humanitarian financing, direct human need, severity, vulnerability, outcomes, or donor intent. A country with low reported coverage may reflect a real financial shortfall, incomplete source reporting, differences in plan structure, or a combination of these factors.

The country lookup improves readability and geographic identification but does not create new funding evidence. It maps codes to names and geographic groupings. Unmatched or non-standard codes should remain visible in data-quality review rather than being assigned a fabricated country identity.

1.10 Limitations

The dashboard is descriptive and should not be used to infer causality. It shows how reported funding compares with recorded requirements in the selected extract. It does not establish why a requirement is underfunded, whether funding arrived on time, whether the reported amount was fully disbursed, or whether a financial gap corresponds to a particular humanitarian outcome.

The annual reporting structure limits the analysis to year-level comparisons. There is no daily or monthly date dimension in the final model because the current question does not require daily time intelligence. A future dashboard involving day, month, quarter, fiscal calendars, YTD, prior-period, or rolling-period analysis would require an explicit date dimension.

The country–plan key preserves the plan-level grain, but plan definitions and reporting practices can vary across countries and years. The existence of a plan record does not guarantee that every requirement or funding field is complete. The source percentage-funded field can also be blank or inconsistent in incomplete records, which is why the dashboard recalculates coverage from the additive amounts.

1.11 Reproducibility checklist

A reproducible project should preserve the downloaded requirements CSV, the country-code reference file, the staging and reference Power Query steps, the final query names, the model relationships, the DAX

measure definitions, the report page layout, the selected source wording, and the project decision log. The Power BI file should be refreshed after any query or relationship change, and the four headline measures should be checked under both the all-period and selected-year contexts.

The final portfolio package should contain the Power BI report, this README, the decision log, the DAX measure catalog, and the source or lookup files used to reproduce the model. The README should be updated whenever the user changes the active query names, analytical grain, measure definitions, or report scope.

1.12 References

1.13 Status

The Executive Overview visual narrative, four KPI cards, annual requirements-and-coverage chart, coverage Shape Map, Top 10 shortfall view, and source/header treatment have been confirmed by the project owner. The active model is requirements-only, and the README now reflects that final analytical scope.